Case Study-1

Scenario: XYZ Investments is evaluating two stocks, A and B, for potential inclusion in their portfolio. They are considering factors like expected return and risk to make a decision. To do this, they are analyzing the historical returns of both stocks over the past five years.

Year	Stock A Return (%)	Stock B Return (%)
1	10%	15%
2	12%	8%
3	8%	12%
4	15%	18%
5	9%	5%

XYZ Investments has a risk tolerance that allows for moderate risk, but they want to ensure that the stock they choose offers the highest **return-to-risk ratio**. Let's evaluate which stock is the better option based on the **expected return** and **risk** (measured by **standard deviation**).

Explanation of Risk and Return

In finance, **return** refers to the profit or loss made on an investment, while **risk** refers to the uncertainty or volatility of returns. Higher returns are typically associated with higher risks. To make an informed investment decision, it's crucial to analyze both aspects together.

- **Expected Return** is the average return an investor anticipates over time.
- **Risk** is often measured by **standard deviation**, which shows how much the returns deviate from the average return (volatility). A higher standard deviation means more risk.

Steps to Solve the Problem:

1. Calculate Expected Return:

The expected return for each stock can be calculated as the average of the historical returns over the 5 years. The formula is:

$$\text{Expected Return} = \frac{\sum \textbf{Returns in Each Year}}{N}$$

Where:

- N is the number of years.

2. Calculate Risk (Standard Deviation):

The standard deviation measures the dispersion of returns from the expected return. The formula for standard deviation is:

$$\sigma 1 = \sqrt{\sum(\text{Return in each year} - \text{Expected Return})2/N}$$

3. Compare the Return-to-Risk Ratio (Sharpe Ratio):

The Sharpe Ratio is a way to understand how much return an investor is getting for each unit of risk. A higher ratio is better, indicating more return for the risk taken.

$$\textit{Sharpe Ratio} = \frac{(Expected\ Return - Risk\ free\ rate)}{Standard\ Deviation}$$

For simplicity, we'll assume the risk-free rate is 0%.

Solution:

1. Calculate Expected Return for Stock A:

Expected Return for Stock A$= \frac{10\%+12\%+8\%+15\%+9\%}{5} = 10.8\%$

Expected Return for Stock B$= \frac{15\%+8\%+12\%+18\%+5\%}{5} = 11.6\%$

2. Calculate Standard Deviation (Risk):

For Stock A,

Calculate the variance first by subtracting each return from the average (10.8%) and squaring the result, then find the average of those squared deviations:

$$\sigma^2_A = \frac{\sum(\textit{Return in each year} - \textit{Expected Return})2}{N}$$

$$= \frac{(10-10.8)^2+(8-10.8)^2+(15-10.8)^2+(9-10.8)^2}{5}$$

=5.36

The standard deviation for Stock A is:

Standard Deviation for Stock $\sigma_A = \sqrt{5.36}$

$= 2.32\%$

For Stock B:

Variance for Stock B $= \frac{(15-11.6)^2+(8-11.6)^2+(12-11.6)^2+(18-11.6)^2+(5-11.6)^2}{5}$

The standard deviation for Stock B is:

Standard Deviation for Stock B= $\sigma_B = \sqrt{19.04}$

=4.36%

3. Calculate Sharpe Ratios:

Since the risk-free rate is assumed to be 0%, the Sharpe Ratio for each stock is:

For Stock A:

Sharpe Ratio for Stock A $= \frac{10.8}{2.32}$

$= 4.66$

For Stock B:

Sharpe Ratio for Stock B $= \frac{11.6}{4.36}$

$= 2.66$

Conclusion:

Based on the Sharpe Ratios, **Stock A** has a higher ratio (4.66) compared to **Stock B** (2.66). This means that for every unit of risk taken, Stock A offers a better return.

While Stock B has a higher expected return (11.6% vs. 10.8%), it also has significantly higher risk, as indicated by its higher standard deviation (4.36% vs. 2.32%). Therefore, for XYZ Investments, **Stock A** would be the better option, especially if they want to maximize returns while keeping risk under control.

Key Takeaways:

- **Risk and Return** are inherently linked in finance: higher potential returns come with higher risk.
- **Expected Return** gives an idea of future earnings, while **Standard Deviation** measures risk.

- The **Sharpe Ratio** is an effective tool for comparing different investment options, helping investors understand how much return they're getting for each unit of risk.

This case study highlights the importance of considering both risk and return when making investment decisions, rather than focusing solely on potential gains.